Process & Decision Documentation

Project/Assignment Decisions

A key design decision was introducing the goal of unlocking a treasure chest. By framing the player as an explorer who must collect keys before unlocking the chest, I aimed to create a simple narrative arc that makes the gameplay feel more purposeful, rewarding, and engaging.

Role-Based Process Evidence

Brainstorming & Creating

Name: Aliza Lallani
Role(s): Sole contributor

Goal of Work Session

Working with GenAI to program the game.

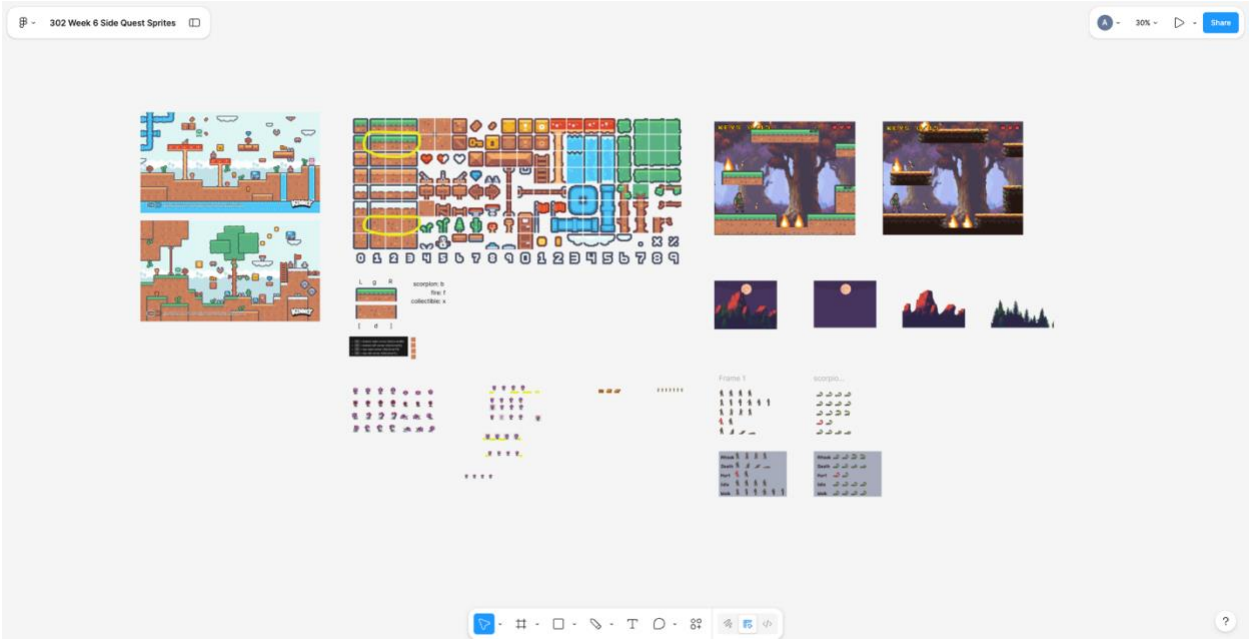
Tools, Resources, or Inputs Used

- Claude Sonnet 4.6 (via website)
- Claude Haiku and Opus 4.5 (via Visual Studio Code)
- Figma

Figma Board

I used Figma as a reference sheet while exploring game ideas and development. This workflow made it easier to choose which sprites to use, adjust their size and arrangement, and plan the overall layout of the game.

Photo on next page



GenAI Documentation

Date Used: March 4th-9th, 2026

Tool Disclosure: Claude Sonnet 4.6 (via website), Claude Haiku and Opus 4.5 (via Visual Studio Code)

Purpose of Use: Guidance on coding and modifying sprite sheets.

Summary of Interaction: The tools helped me edit Example 5 to transform my game idea into specific code.

Human Decision Point(s): I reviewed each GenAI suggestion, either accepting and implementing it, requesting specific modifications, or overruling it and deciding on a different path. Specific examples are noted below under *Decision Points & Trade-offs*.

Integrity & Verification Note: I reviewed the syllabus to ensure my GenAI use was appropriate and treated GenAI as a collaborative partner rather than a substitute, iterating until I achieved my desired output.

Scope of GenAI Use: GenAI was used to generate specific code snippets and clarify where to insert them. GenAI also helped me learn how sprite sheets can be modified to fit certain dimensions. However, I maintained creative control, critically reviewing and refining the AI suggestions to fit my design goals. I also came up with the initial idea for the game on my own.

Limitations or Misfires: There were several instances where GenAI's code didn't behave as I expected or broke the game, particularly when debugging and working to add another level.

Summary of Process (Human + Tool)

- *[Human]* Brainstormed game ideas
- *[Human]* Prompt engineering
- *[Human + AI]* Adjusted Example code to match my chosen game idea
- *[Human]* Used Figma to edit sprite sheet layouts and sizing for key, treasure chest, plant, and tiles.
- *[Human]* Adjusted tilemap to change game layout
- *[Human]* Consulted Prof David
- *[Human]* Tested code and adjusted details

Decision Points & Trade-offs

I initially planned to design a space-themed game with an alien as the main character. However, the sprite sheet I downloaded included multiple characters and was not clearly separated. I attempted to reorganize it in Figma and tested several versions in the code, but ultimately decided it would be more practical to switch to a different theme and sprite sheet. Despite this change, the process helped me better understand how sprite sheets are organized and referenced in code, which made working with other sprite sheets easier.

Verification & Judgement

With each change, I reviewed the project in the "Live Server" view and evaluated whether it supported the gameplay or detracted from it. I also iterated extensively with GenAI, refining or rejecting outputs that didn't align with my creative vision.

Appendix

Claude Sonnet 4.6

https://claude.ai/share/99ea4410-12b4-49cf-a7cf-e72ae695c37b

Claude Haiku 4.5

Unfortunately, this chat does not appear in my chat history. I've tried multiple methods to locate it without success. For transparency, I've included a list of the tasks I used this model for:

- Adjusting code to accommodate new tile sprite sheet

- Adjusting code to accommodate new plant sprite sheet
- Learning how to modify tilemap
- Adjusting code for treasure chest functionality
- Adding popup messages when player reaches treasure chest
- Fixing a bug where more scorpion enemies spawned than intended
- Adding sounds for jumping, attacking, taking damage, and collecting objects
- Adding looping background music

Claude Opus 4.5

Please see the following pages.

Button width adjustment request — alallani_sidequest_W6

Increase width of Continue Exploring and Play Again Buttons

✓ Reviewed FlagPopup.js lines 1 to 201

I can see the buttons are currently set to a width of 100. I'll increase this to make them wider. Let me update both the draw method and the handleClick method where the button width is defined.

✓ Updated FlagPopup.js

Done! I've increased the button width from 100 to 140 in both the `draw()` and `handleClick()` methods. The buttons will now be wider while remaining properly centered in the popup.

Increase popup width to 210

✓ Updated FlagPopup.js

Done! The popup width is now 210 in both the `draw()` and `handleClick()` methods.

When popups are on screen, player should not be able to control character. Only interaction is pressing the Play Again or Continue Exploring buttons

✓ Reviewed 5 files and searched for 2 terms

I'll modify the Game class to accept the flagPopup as an option and disable player controls when it's active. Let me first check the current structure and then make the changes.

✓ Reviewed and updated Game.js and main.js

Done! I've implemented the feature to disable player controls when the FlagPopup is active. Here's what I changed:

1. **Modified** `Game.js` :
   - Added `flagPopup` to the constructor options
   - Added logic in the `update()` method to clear movement and action inputs (left, right, jump, attack) when the popup is active

> 3 files changed +43 -7 Keep Undo

+ JS p5play.min.js

Describe what to build next

Local Agent Claude Haiku 4.5

Button width adjustment request — alallani_sidequest_W6

1. **Modified** Game.js :
   - Added flagPopup to the constructor options
   - Added logic in the update() method to clear movement and action inputs (left, right, jump, attack) when the popup is active
2. **Modified** main.js :
   - Reordered the code to create UI overlays (including flagPopup) before initializing Game
   - Added flagPopup to the Game constructor options

Now when the popup appears, players can't move or jump—they can only interact with the "Play Again" or "Continue Exploring" buttons. Other inputs like restart aren't affected by the popup.

Clicking the Play Again button should restart the game

✓ Reviewed Level.js and FlagPopup.js, updated main.js

Done! I've implemented the feature so clicking "Play Again" restarts the game. Here's what I changed:

1. **Modified** FlagPopup.js :
   - Added lastAction property to track which button was clicked
   - Updated handleClick() to record the action ("playAgain" or "continueExploring") before hiding
   - Added getAction() method to retrieve and clear the action
   - Updated hide() to also clear the action
2. **Modified** main.js :
   - Added code after game.update() to check for popup actions
   - When "playAgain" action is detected, calls game.level.restart()

Now when the player clicks "Play Again" on the treasure unlocked screen, the level will restart.

Restore Checkpoint

Code no longer loads

✓ Reviewed FlagPopup.js and updated it

Fixed! The issue was a missing closing brace for the FlagPopup class. I've added it back and the code should now load properly.

3 files changed +43 -7 Keep Undo

p5play.min.js

Describe what to build next

Local Agent Claude Haiku 4.5

Button width adjustment request — alallani_sidequest_W6

Play Again button needs to restart game - this currently doesn't work

✓ Reviewed 5 files and searched for level:restarted

Let me check the actual button detection and restart flow more carefully:

✓ Reviewed FlagPopup.js lines 182 to 209

I see the issue - the file has a syntax error with double closing braces. Let me also verify the restart logic is being called properly by adding better integration between the popup and game restart:

✓ Reviewed Game.js lines 1 to 30

Let me add a restart method to the Game class for better control:

✓ Searched for draw function and modified Game.js and FlagPopup.js

Done! I've made the following fixes to get the Play Again button working:

1. **Added a** `restart()` **method to** `Game.js` - This properly encapsulates the restart logic and ensures the Game's event listeners handle state cleanup correctly.
2. **Updated** `main.js` - Changed to call `game.restart()` instead of directly calling `game.level.restart()`.
3. **Fixed** `FlagPopup.js` - Removed the syntax error with the double closing braces.

The restart flow now works as:

- User clicks "Play Again" button
- `handleClick()` records the action and hides the popup
- Next frame, `game.restart()` is called, which:
  - Calls `level.restart()` to reset all game entities
  - Triggers the "level:restarted" event, which resets Game's win/lost flags and other state via the event listener

The game should now properly restart when "Play Again" is clicked.

It still doesn't work - Play Again does the same thing as Continue Exploring. Each button should have distinct functions:

Continue Exploring – Return to game (currently works)

3 files changed +43 -7 Keep Undo

p5play.min.js

Describe what to build next

Local Agent Claude Haiku 4.5

Button width adjustment request — alallani_sidequest_W6

It still doesn't work - Play Again does the same thing as Continue Exploring. Each button should have distinct functions:

Continue Exploring = Return to game (currently works)
Play Again = completely restart game (currently does not work)

FlagPopup.js

✓ Analyzed Play Again button functionality and restart logic

I found the bug! In `FlagPopup.hide()`, we're clearing `lastAction`, but that happens before we get a chance to read it in the next frame.

✓ Updated FlagPopup.js

Done! The bug was that `hide()` was clearing `lastAction` immediately, but we needed to keep it until the next frame when `getAction()` could read it.

The sequence now works correctly:

1. Player clicks the button
2. `handleClick()` records the action (`"playAgain"` or `"continueExploring"`) and calls `hide()`
3. Next frame, `getAction()` reads and clears the action
4. If it was `"playAgain"`, `game.restart()` is called to restart the level
5. If it was `"continueExploring"`, nothing happens and the game continues

The "Play Again" button should now properly restart the game.

Play Again button does restart the game now, but the flag is stuck on its final state. The flag and popups should restart as well.

✓ Searched for text `flag`, reviewed 3 files, updated Level.js

Done! I've added flag reset code to the `restart()` method in Level.js. When the game restarts:

1. The flag sprite's `_waveTriggered` property is reset to `false`
2. The flag animation is changed back to `"idle"`
3. The "level:restarted" event is emitted, which triggers the listener in `main.js` to hide the popup

Now when you click "Play Again", the flag will reset to its initial state along with all other game entities, and the popup will be hidden.

3 files changed +43 -7 Keep Undo

p5play.min.js

Describe what to build next

Local Agent Claude Haiku 4.5

Replacing background assets in project — alallani_sidequest_W6

I've added the following assets:
skyBackground1.png
mountainsBackground2.png
treesBackground3.png

I want them to replace the following in order:
background_layer_1.png
background_layer_2.png
background_layer_3.png

✓ Searched for background_layer and updated levels.json

Replaced all 6 references:

- `background_layer_1.png` → `skyBackground1.png`
- `background_layer_2.png` → `mountainsBackground2.png`
- `background_layer_3.png` → `treesBackground3.png`

Updated in `levels.json:46-48` and `AssetLoader.js:151-153`.

The parallax effect isn't working for the background

✓ Reviewed 4 files and searched for 4 terms

Let me check the CameraController to see how camera.x is being set:

✓ Reviewed 5 files and searched for 2 terms

I found the bug. The parallax tiling logic quantizes to full tile boundaries, losing the smooth pixel-level offset. When the camera moves, the background should shift smoothly, but currently it only "jumps" at tile boundaries.

✓ Updated ParallaxBackground.js

The fix addresses two issues:

1. Added `!img.width` check to skip unloaded images (prevents potential issues with async loading)
2. Fixed the tiling offset calculation to preserve the smooth pixel-level shift instead of quantizing to tile boundaries

The parallax should now smoothly scroll as the camera moves. Each layer will move at its configured speed (0.2, 0.4, 0.6) relative to camera movement.

p5play.min.js

Describe what to build next

Local · Agent · Claude Opus 4.5

Replacing background assets in project — alallani_sidequest_W6

The text colour for "Deep in the forest lies hidden treasure, locked tight. Can you find all the keys to claim it, or will sneaky scorpions and poisonous plants get in your way?" is still black. Change it to WHITE

✓ Reviewed index.html and searched for style.css

The `style.css` file isn't linked in the HTML. I'll add the link and also apply the white color inline to ensure it works:

✓ Updated index.html

Done. Added the stylesheet link and applied `style="color: white;"` directly inline on the text to ensure it displays white.

Add a start button below the text. The game should only start when player presses that button.

✓ Reviewed index.html and style.css, updated index.html

Done. Added a "Start" button below the text. The intro screen now only dismisses when the player clicks the button, not anywhere on the screen.

Add information for controls:
WASD/Arrows to move
SPACE to attack

✓ Reviewed and updated index.html and style.css

Done. Added controls info between the description and the start button, styled in a slightly subdued white.

Make it a bit larger

✓ Updated style.css

Done. Increased the controls text size from 1rem to 1.3rem

JS p5play.min.js

Describe what to build next

Local Agent Claude Opus 4.5

Replacing background assets in project — alallani_sidequest_W6

Use purpleBackground1.png instead of sky1.png

✓ Updated levels.json and searched for sky1.png and skyBackground1.png

Done. Replaced `skyBackground1.png` with `purpleBackground1.png` in levels.json:46 and AssetLoader.js:151.

I want to add this intro text, either in HTML or as a start screen within the game: Deep in the forest lies hidden treasure, locked tight. Can you find all the keys to claim it, or will sneaky scorpions and poisonous plants get in your way?

✓ Reviewed 3 files and updated index.html

Added an intro screen overlay with your text and a "Start Game" button. The overlay fades out when clicked, revealing the game underneath. The styling uses a forest-themed color scheme that matches the game's atmosphere.

Make the text larger and a white color. Rremove "Start Game" button

✓ Reviewed 2 files and updated index.html and style.css

Done. Text is now larger (1.6rem) and white. Removed the button — clicking anywhere on the intro screen now dismisses it.

The text color is still black - it needs to be white. Also, we don't need any text or buttons prompting the player to start - the description text is enough.

✓ Reviewed index.html and updated style.css

Done. Removed the "Click anywhere to start" hint and fixed the text color to white with `!important` to ensure it's applied.

p5play.min.js

Describe what to build next

Local Agent Claude Opus 4.5

Replacing background assets in project — alallani_sidequest_W6

Make it a bit larger

✓ Updated style.css

Done. Increased the controls text size from 1rem to 1.3rem.

The player should not be able to control the character before pressing "Start Exploring" button

✓ Reviewed 3 files, updated InputManager.js, and searched for terms

Done. Added a `window.gameStarted` flag that starts as `false`. The InputManager now checks this flag and returns neutral (no input) until the "Start Exploring" button is pressed, which sets `window.gameStarted = true`.

Start background music loop only when player presses "Start Exploring"

✓ Reviewed main.js and updated it

Done. The background music will now only start after the player clicks "Start Exploring".

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flagSpriteSheet.png has been renamed to chestSpriteSheet.png - adjust code accordingly

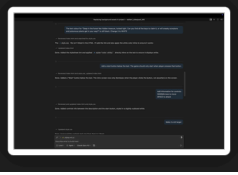
✓ Searched for flagSpriteSheet.png and updated AssetLoader.js

Done. Updated `flagSpriteSheet.png` to `chestSpriteSheet.png` in `AssetLoader.js:38` and `README.md:23`.

p5play.min.js

Describe what to build next

Local · Agent · Claude Opus 4.5



MIME type error in GitHub Pages deployment — alallani_sidequest_W6

The code appears to be running in Live Server, but not in GitHub. The console shows this error message: 10:04:00.338 Loading module from “https://alallani.github.io/alallani_sidequest_W6/src/world/TileBuilder.js” was blocked because of a disallowed MIME type (“text/html”). alallani_sidequest_W6

✓ Tried to run code on GitHub Pages but was blocked

The issue is a case sensitivity problem. Looking at your workspace structure, the file is named `TIleBuilder.js` (capital I in "TIle"), but the import references `TileBuilder.js` (lowercase 'i'). This works on macOS (case-insensitive) but fails on Linux servers like GitHub Pages (case-sensitive).

Let me find and fix the import:

✓ Searched for text TileBuilder

Found it. The file is named `TIleBuilder.js` (typo with capital 'I') but imported as `TileBuilder.js`. I'll rename the file to fix this:

✓ Renamed file and planned commit to GitHub

Done. The file has been renamed from `TIleBuilder.js` to `TileBuilder.js`.

Commit and push this change to GitHub:

```
git add -A && git commit -m "Fix TileBuilder.js filename case" && git push
```

The case mismatch caused GitHub Pages (Linux, case-sensitive) to return a 404 HTML page instead of the JS file, hence the MIME type error.

There appears to be a problem when running the game in Live Server now - everything is slower and the player looks like it's floating when jumping compared to before

✓ Analyzed physics and player movement system issues

Let me investigate the player physics and game timing:

✓ Reviewed 5 files and checked workspace for problems

The file rename I made shouldn't affect game behavior on Live Server - macOS is case-insensitive, so it was loading correctly locally before.

Could you clarify: Was the game working correctly **before** you noticed the GitHub error? Or is this "slower/floating" behavior something you've been seeing all along?

Add Context...

Describe what to build next

Local · Agent · Claude Opus 4.5

Also, to properly fix the case sensitivity issue for GitHub, the rename needs to be staged properly since macOS doesn't register case-only changes. Here's how to commit it:

```
git mv src/world/TIleBuilder.js src/world/tilebuilder_temp.js
git mv src/world/tilebuilder_temp.js src/world/TileBuilder.js
git commit -m "Fix TileBuilder.js filename case"
git push
```

The game in Live Server was working correctly before our chat.

✓ Considered file renaming impact on macOS case sensitivity

Let me undo the rename I made - that should restore the original working state:

✓ Restored file name and planned GitHub fix.

Done - I've restored the original filename. Try refreshing Live Server now.

My rename was the issue. On macOS, file case changes can sometimes trigger the file watcher in unexpected ways, potentially causing p5play to reinitialize or lose state.

To fix the GitHub Pages issue properly (without breaking local), we need to rename the file in git correctly:

```
git mv src/world/TIleBuilder.js src/world/temp_tilebuilder.js
git mv src/world/temp_tilebuilder.js src/world/TileBuilder.js
git commit -m "Fix TileBuilder.js filename case for GitHub Pages"
git push
```

This two-step rename forces git to track the case change. Let me know once you've confirmed Live Server is working again, and I can run these

Code in Live Server is still not right. Restore the version before our chat.

✓ Restored everything to last committed state and checked files

All files restored to the last committed state. Try refreshing Live Server now.

The working tree is clean and matches what's in git. If the issue persists, it may be something unrelated to file changes - perhaps a browser cache issue (try Cmd+Shift+R for hard refresh).

Add Context...

Describe what to build next

Local · Agent · Claude Opus 4.5

MIME type error in GitHub Pages deployment — alallani_sidequest_W6

The issue persists in my normal browser but I opened Live Server in Incognito mode and my code works as normal

✓ Confirmed browser cache issue affecting code functionality

It's a browser cache issue. Clear your browser cache for localhost:

Chrome/Edge:

1. Open DevTools (Cmd+Option+I)
2. Right-click the refresh button → "Empty Cache and Hard Reload"

Or: Go to Settings → Privacy → Clear browsing data → select "Cached images and files" → Clear

After clearing, your normal browser should work correctly again.

To fix the GitHub Pages issue (the original case-sensitivity problem), run:

```
git mv src/world/TIleBuilder.js src/world/temp_tilebuilder.js && git mv src/world/temp_tilebuilder.js src/world/TileBuilder.j
```

Want me to run this for you?

Yes

✓ Renamed TIleBuilder.js to TileBuilder.js and pushed changes

Done. The fix has been pushed to GitHub. Give GitHub Pages a minute to rebuild, then your game should work at `https://alallani.github.io/alallani_sidequest_W6/`.

Add Context...

Describe what to build next

Local · Agent · Claude Opus 4.5